THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 14: **Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An L-shaped figure is made by removing a 8 cm by 6 cm rectangle from a 20 cm by 15 cm rectangle. Its area is: 1

(a) 300 cm² **(b)** 348 cm² **(c)** 252 cm² **(d)** 108 cm²

2. A parallelogram has base 18 cm and height 10 cm. If the base is changed to 15 cm (same area, other side), the corresponding height (in cm) is: 1

(a) 15 **(b)** 12 **(c)** 13 **(d)** 165

3. A rhombus has side 15 cm and one diagonal 18 cm. Its area is: 1

(a) 60 cm² **(b)** 216 cm² **(c)** 270 cm² **(d)** 108 cm²

4. A path 2 m wide runs around the outside of a 20 m by 15 m rectangular garden. The area of the path is: 1

(a) 140 m² **(b)** 156 m² **(c)** 16 m² **(d)** 456 m²

5. A right triangle has hypotenuse 25 cm and one leg 15 cm. Its area is: 1

(a) 250 cm² **(b)** 300 cm² **(c)** 187 cm² **(d)** 150 cm²

6. The diagonal of a square is 10 cm. Its area is: 1

(a) 20 cm² **(b)** 100 cm² **(c)** 25 cm² **(d)** 50 cm²

7. An L-shaped figure is made by removing a 10 cm by 7 cm rectangle from a 25 cm by 18 cm rectangle. Its area is: 1

(a) 450 cm² **(b)** 520 cm² **(c)** 165 cm² **(d)** 380 cm²

8. A parallelogram has base 20 cm and height 12 cm. If the base is changed to 16 cm (same area, other side), the corresponding height (in cm) is: 1

(a) 14 **(b)** 15 **(c)** 18 **(d)** 224

9. A rhombus has side 26 cm and one diagonal 20 cm. Its area is: 1

(a) 104 cm² **(b)** 520 cm² **(c)** 240 cm² **(d)** 480 cm²

10. A path 3 m wide runs around the outside of a 30 m by 18 m rectangular garden. The area of the path is: 1

(a) 864 m² **(b)** 324 m² **(c)** 288 m² **(d)** 36 m²

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A rectangle has perimeter 80 cm and length 26 cm. Find its breadth and area. 2

12. A rhombus has side 50 cm and one diagonal 28 cm. Find its other diagonal and area. 2

13. A square garden has a diagonal of 18 m. Find its area and side length (leave the side as a surd). 2

14. A rectangle has perimeter 100 cm and length 31 cm. Find its breadth and area. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rectangular park is 40 m by 30 m. A path 2 m wide runs inside along its boundary. Find the area of the path. 3

16. A parallelogram has base 24 cm and height 10 cm. Another side of 16 cm has a corresponding height; find that height and the area. 3

17. A right triangle has legs 9 cm and 12 cm. Find its hypotenuse and area, and the length of the altitude drawn to the hypotenuse. 3

OR

A rectangular park is 50 m by 36 m. A path 3 m wide runs inside along its boundary. Find the area of the path. 3

18. A parallelogram has base 30 cm and height 12 cm. Another side of 18 cm has a corresponding height; find that height and the area. 3

OR

A right triangle has legs 15 cm and 8 cm. Find its hypotenuse and area, and the length of the altitude drawn to the hypotenuse. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A figure is made of a rectangle 20 cm by 10 cm with a right triangle of base 20 cm and height 8 cm placed on one of its longer sides. Find the total area and the area of the triangle as a percentage of the total. 5

OR

The parallel sides of a trapezium are 32 cm and 18 cm and its area is 250 cm². Find the distance between the parallel sides. A parallelogram of base 18 cm (the shorter parallel side) and the same height is cut out, leaving a triangle; find the area of the parallelogram and of the triangle left. 5

20. A figure is made of a rectangle 24 cm by 12 cm with a right triangle of base 24 cm and height 10 cm placed on one of its longer sides. Find the total area and the area of the triangle as a percentage of the total. 5

OR

The parallel sides of a trapezium are 28 cm and 14 cm and its area is 252 cm². Find the distance between the parallel sides. A parallelogram of base 14 cm (the shorter parallel side) and the same height is cut out, leaving a triangle; find the area of the parallelogram and of the triangle left. 5

END OF QUESTION PAPER — Total: 40 Marks